

Superstore Sales Analysis & Forercasting

- Power BI Project



Project Goal

The primary objective of this project is to analyze historical Superstore sales data (2018–2021) to:

- Understand overall business performance and growth trends
- Identify key sales drivers across time, region, customer, and product dimensions
- Detect loss-making products and optimization opportunities
- Forecast next year's sales using data-driven methods
- Deliver actionable insights for strategic decision-making

PESTEL Analysis

Political

Retail sales are influenced by regional tax structures, trade regulations, and interstate commerce policies.

A stable political environment supports long-term operational planning and store expansion.

Technological

Advanced analytics and visualization tools enable real-time performance monitoring and forecasting.

Data-driven decision-making improves pricing, inventory planning, and demand prediction.

Economic

Macroeconomic factors such as inflation, consumer purchasing power, and economic recovery cycles directly impact sales volume.

The observed post-2019 recovery suggests resilience and growth potential under stable economic conditions.

Environmental

Growing awareness of sustainability impacts packaging, logistics, and supplier choices.

Operational efficiency can reduce both environmental footprint and cost.

Social

Changing consumer behavior, increased online purchasing, and demand for fast delivery affect product mix and sales strategy.

A high proportion of repeat customers indicates strong brand loyalty and customer satisfaction.

Legal

Compliance with consumer protection laws, pricing transparency, and return policies is essential.

Accurate reporting and discount management reduce legal and reputational risks.

Fact_OrderLine	...
City	
Country/Region	
Σ Discount	
Σ Grossprofit	
Σ GrossSale	
NetSale	
Order ID	
OrderlineID	
Product ID	
Σ Quantity	
Region	
State	
UnitDiscount	
UnitPrice	
Collapse ^	

Dim_Customer	...
City	
Country	
Customer ID	
Customer Name	
Postal Code	
Region	
Segment	
State	
Collapse ^	

Fact_order	...
Customer ID	
Datekey	
Order Date	
Order ID	
Ship Date	
Ship Mode	
Collapse ^	

Dim_Date	...
Datekey	
Day Name	
Σ Day of Week	
Month Name	
Σ Monthnumber	
Order Date	
Quarter	
Σ Quarternumber	
Σ Week of Month	
Σ Year	
Collapse ^	

Dim_Product	...
Category	
Product ID	
Product Name	
SubCatID	
Collapse ^	

SubCategory	...
Categoryid	
Sub-Category	
SubCatID	
Sales MoM	
Collapse ^	

Top N repeated Products	...
Top N repeated Products	
Top N repeated Products Value	
Collapse ^	

Category	...
Category	
Categoryid	
Collapse ^	

Executive overview

Business performance shows a strong recovery and growth trend.

Total sales reached 2.30M with a net profit of 286,40K, driven primarily by repeat customers (98%) and strong performance in the West and

East regions.

Sales declined in 2019 but recovered significantly in 2020 and peaked in 2021, indicating a resilient and scalable business model.

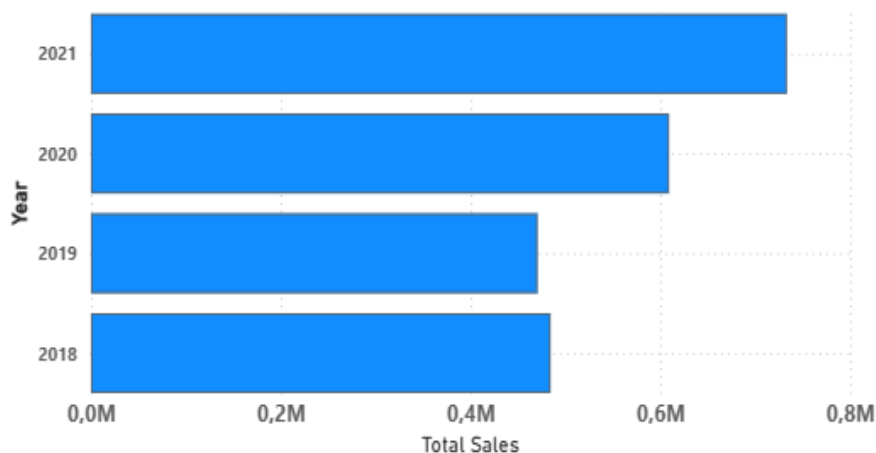
However, 301 products are loss-making, generating a total loss of 76.7K, highlighting the need for product and pricing optimization.

Overall, the business demonstrates solid growth potential, with clear opportunities to improve profitability by addressing underperforming products and focusing on high-performing regions.

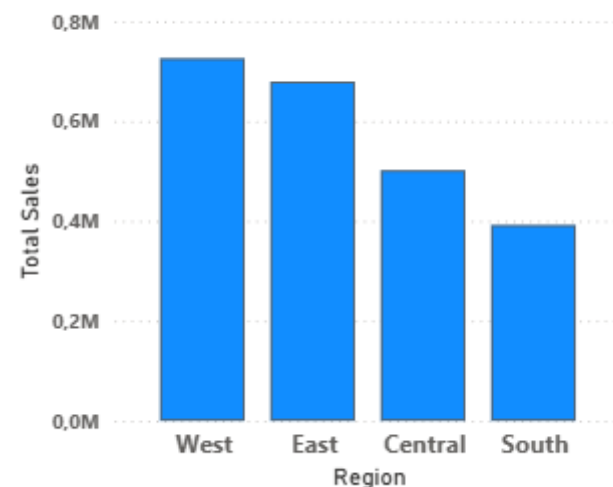
2,30M	1,56K	2,30M	286,40K
Total Sales	Total Discount	Net Sales	Total Profit
38K	60,66	0,04	0,12
Total Quantitv	Ava Unit Price	Ava Discount per Unit	Margin %
793	12	781	0,98
Total Customers	One-time Customers	Repeat Customers	Repeat Customer %

1862	1558	301	3	-76,72K
TotalProducts	Profitable Products	LossProducts	product Not sale	Total Loss Amount

Total Sales by Year



Total Sales by Region



Sales Performance



Time-Based Sales Performance Overview

This page provides a comprehensive time-based sales analysis, covering yearly, quarterly, monthly, and day-level performance to identify trends, seasonality, and demand patterns.

At the year level, sales demonstrate a consistent upward trend, reflecting overall business growth and improved performance over time.

Quarter-wise analysis highlights stronger results in the second half of the year, with Q4 emerging as the peak revenue period, driven by higher customer demand and seasonal buying behavior. On a monthly basis, sales remain relatively stable in mid-year but increase sharply toward October–December, indicating strong year-end momentum.

Day-wise analysis reveals that sales are concentrated on weekdays, especially mid-week, suggesting that business demand is primarily driven by operational and corporate purchasing rather than weekend consumer activity. Overall, this time-based view enables stakeholders to optimize planning, promotions, and resource allocation by aligning strategies with high-performing periods and anticipating future demand patterns.

Sales growth is driven by strong Q4 seasonality and consistent weekday demand patterns.

Year

Deselect all

2018

2019

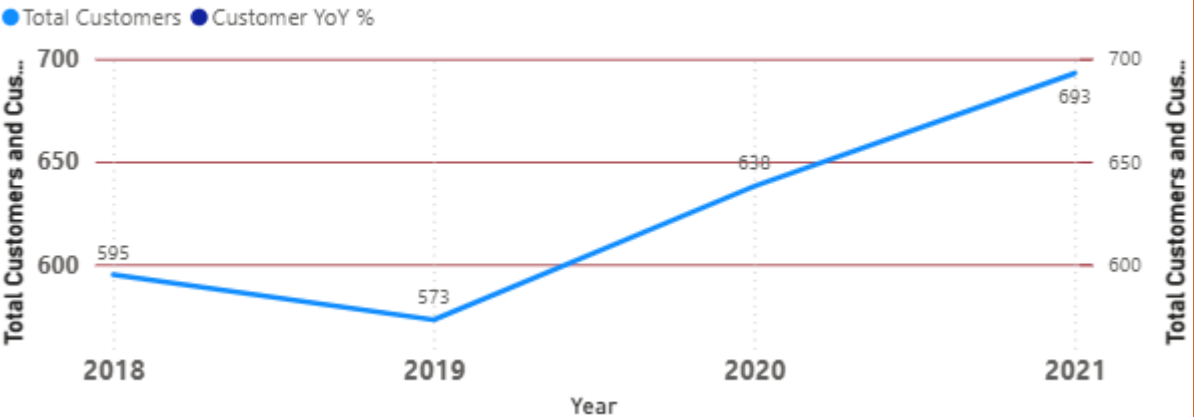
2020

2021

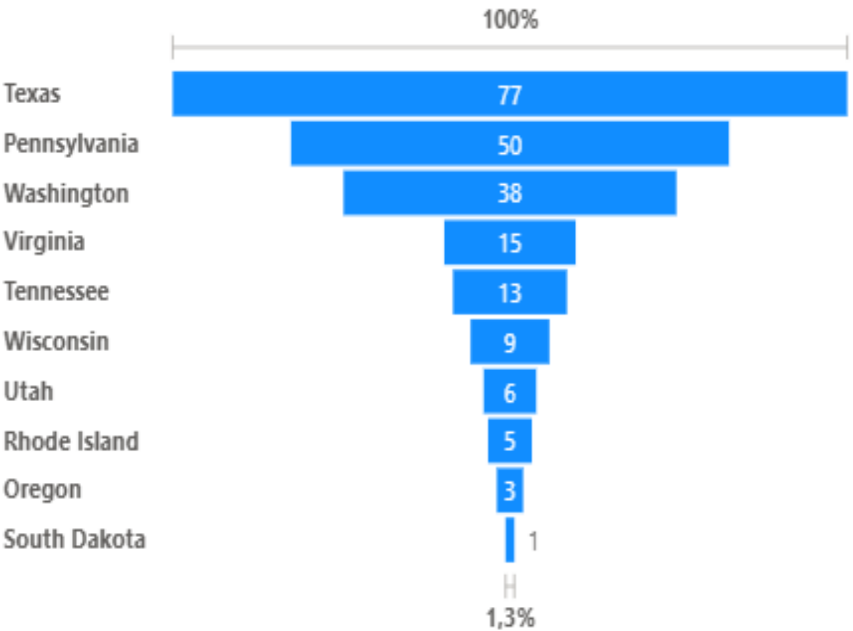
Top 10 Customers

Customer Name	Total Orders	Net Sales
Adrian Barton	10	14.468,77
Christopher Conant	5	12.125,98
Hunter Lopez	6	12.873,10
Ken Lonsdale	12	14.169,43
Raymond Buch	6	15.115,65
Sanjit Chand	9	14.140,94
Sanjit Engle	11	12.207,34
Sean Miller	5	25.039,37
Tamara Chand	5	19.050,82
Tom Ashbrook	4	14.594,82
Total	73	153.786,22

Total Customers and Customer YoY % by Year



Total Customers by State



Customer Performance & Retention Overview

This page highlights customer value, geographic distribution, and retention trends. A small group of top customers generates a significant share of sales, indicating strong high-value relationships. Customer presence is concentrated in key states, while several regions remain under-penetrated, offering expansion opportunities. Year-over-year analysis shows steady customer growth after 2019, reflecting improved acquisition and recovery. With a 98% repeat customer rate, retention is a major strength, demonstrating high customer loyalty and long-term business stability.

793

Total Customers

781

Repeat Customers

12

One-time Customers

0,98

Repeat Customer %

Repeated Products by Slide

17

1862
TotalProducts

17 Times repeated Products

Product Orders Count	Product Name
18	Xerox 1881
17	Storex Dura Pro Binders
19	Staples in misc. colors
46	Staples
18	Staple remover
48	Staple envelope
18	Logitech P710e Mobile Speakerphone
18	KI Adjustable-Height Table
46	Easy-staple paper
18	DAX Wood Document Frame
20	Avery Non-Stick Binders

Top 10 most selling product

Product Name	Sub-Category	Net Sales
Canon imageCLASS 2200 Advanced Copier	Copiers	61.599,23
Cisco TelePresence System EX90 Videoconferencing Unit	Machines	22.637,98
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	Binders	27.450,98
GBC DocuBind P400 Electric Binding System	Binders	17.962,37
GBC DocuBind TL300 Electric Binding System	Binders	19.820,18
GBC Ibimaster 500 Manual ProClick Binding System	Binders	19.019,80
Hewlett Packard LaserJet 3310 Copier	Copiers	18.838,08
High Speed Automatic Electric Letter Opener	Supplies	17.030,11
HON 5400 Series Task Chairs for Big and Tall	Chairs	21.868,97
Total		244.602,10

Top 10 Profitable Product

Product Name	Sub-Category	Total Profit
Zebra ZM400 Thermal Label Printer	Machines	3.343,53
Logitech G19 Programmable Gaming Keyboard	Accessories	4.425,32
Ibico EPK-21 Electric Binding System	Binders	3.345,29
HP Designjet T520 Inkjet Large Format Printer - 24" Color	Machines	4.094,98
Hewlett Packard LaserJet 3310 Copier	Copiers	6.983,89
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	Binders	7.753,06
Canon PC1060 Personal Laser Copier	Copiers	4.570,94
Canon imageCLASS 2200 Advanced Copier	Copiers	25.199,94
Ativa V4110MDD Micro-Cut Shredder	Machines	3.772,95
Total		67.207,87

Top 10 Most Loss Products

Product Name	Sub-Category	Product Loss Amount
Balt Solid Wood Round Tables	Tables	-1.201,06
BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables	Tables	-1.148,44
Bush Advantage Collection Racetrack Conference Table	Tables	-1.934,40
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	Tables	-2.876,11
Cisco TelePresence System EX90 Videoconferencing Unit	Machines	-1.811,08
Cubify CubeX 3D Printer Double Head Print	Machines	-8.879,97
Cubify CubeX 3D Printer Triple Head Print	Machines	-3.839,99
GBC DocuBind P400 Electric Binding System	Binders	-1.878,17
Lexmark MX611dhe Monochrome Laser Printer	Machines	-4.589,97
Martin Yale Chadless Opener Electric Letter Opener	Supplies	-1.299,19
Total		-29.458,38

Product Performance & Profitability Overview

This page provides a comprehensive view of product performance, demand consistency, and profitability across the Superstore portfolio.

Out of 1,862 total products, a clear demand concentration is visible. A subset of products is repeatedly ordered, indicating strong and consistent customer demand. The interactive *Repeated Products* control highlights items ordered multiple times, helping identify core, high-velocity products that drive recurring revenue.

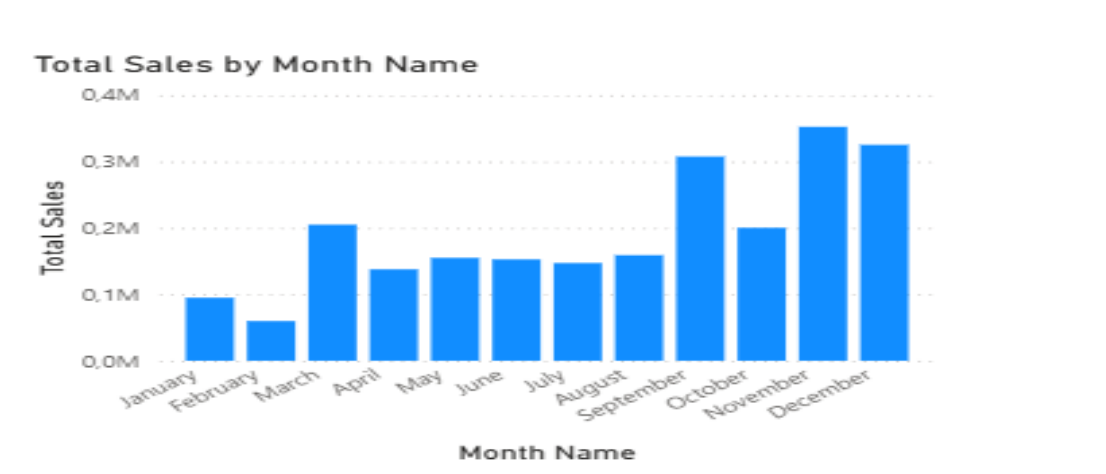
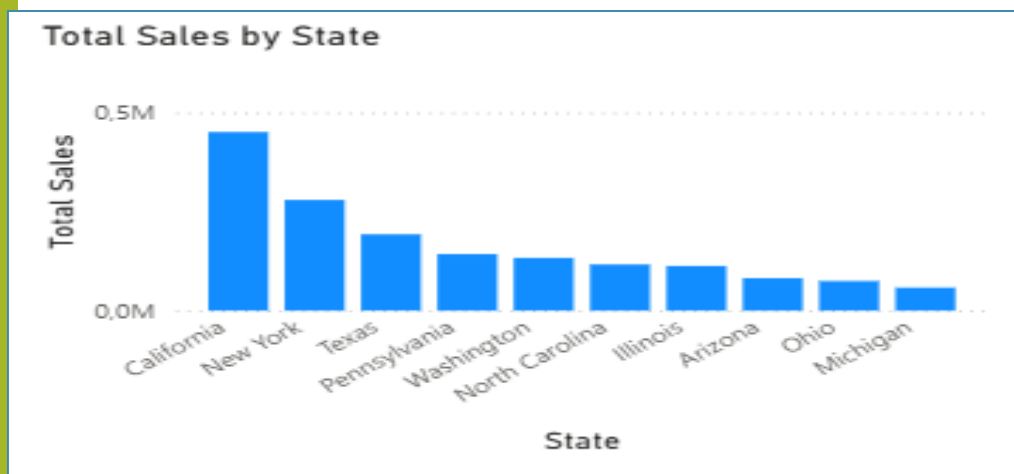
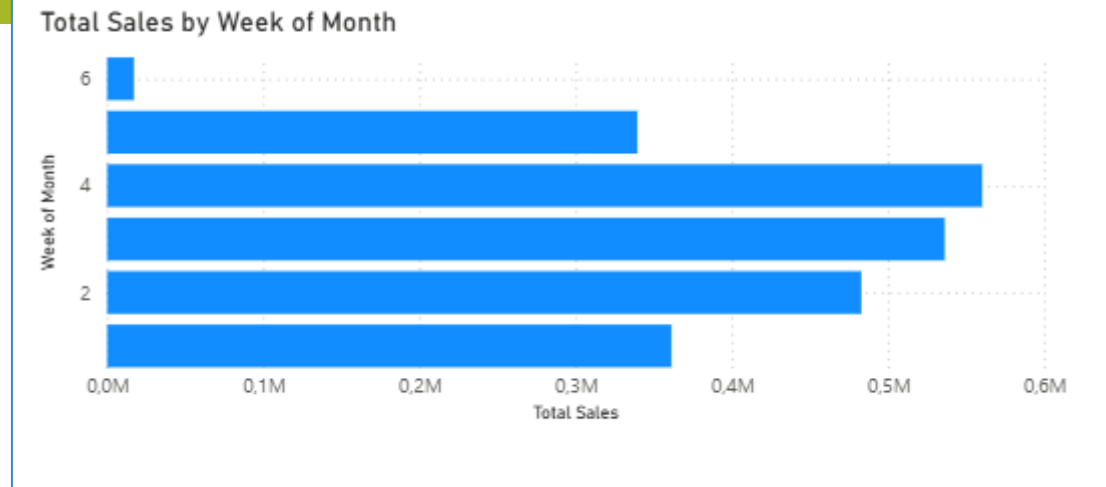
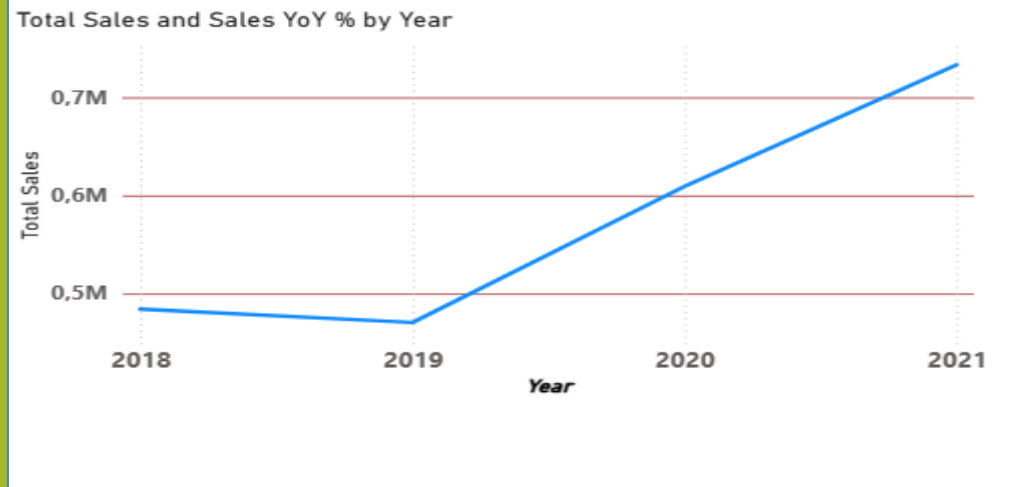
The Top 10 Most Selling Products represent the primary sales contributors, dominated by office equipment and technology categories, while the Top 10 Profitable Products reveal that high revenue does not always equate to high profit—profitability is driven by a mix of pricing, cost structure, and repeat demand.

At the same time, the loss-making products section exposes underperforming items generating negative margins. These products highlight opportunities for pricing optimization, cost reduction, supplier renegotiation, or portfolio rationalization.

Overall, this analysis enables data-driven decisions to:

- Prioritize high-demand and high-profit products
- Reduce exposure to loss-making items
- Strengthen inventory planning based on repeat purchase behavior
- Align product strategy with profitability, not just sales volume

This page supports strategic actions aimed at sustainable growth, margin improvement, and operational efficiency.



0,47
Sales YoY %
733,22K
Last Year Sales
149,25K
Forecast Delta
609,21K
Previous Year Sales
882,47K
Forecast Next Year Sales

Time & Regional Performance Analysis

This section analyzes sales performance across time dimensions (year, quarter, month, and day) and geographic regions to identify growth patterns, seasonality, and regional strengths.

Sales show a consistent upward trend year over year, with strong acceleration in the most recent year, indicating healthy business growth. Monthly analysis reveals clear seasonality, with peak performance during the later months of the year, supporting demand planning and inventory optimization. Day-level patterns further highlight specific weekdays with higher sales activity, enabling better staffing and operational planning.

From a regional perspective, revenue is unevenly distributed, with the West and East regions leading total sales, while Central and South regions show moderate but stable contributions. State-level insights identify top-performing markets and underpenetrated regions, creating opportunities for targeted regional strategies and localized marketing initiatives.

Overall, this analysis helps the business understand when and where sales are generated, enabling more accurate forecasting, regional prioritization, and time-based decision-making.

Business Forecast & Growth Outlook

The business forecast projects next-year sales based on historical performance and year-over-year growth trends.

Using the latest completed year as a baseline, the forecast indicates a strong positive growth trajectory, with expected sales increasing significantly in the upcoming year. The forecast delta highlights incremental revenue potential, validating the effectiveness of current sales momentum and customer retention strategies.

This forward-looking view enables stakeholders to:

- Set realistic revenue targets
- Plan capacity, inventory, and staffing
- Align marketing and regional expansion strategies
- Identify early signals of growth acceleration or slowdown

Overall, the forecast confirms that the business is positioned for continued expansion, provided current performance trends and operational efficiency are maintained.

Business Storytelling (Executive Narrative)

This project provides a comprehensive analysis of Superstore's sales performance over four years. The analysis reveals a temporary slowdown in 2019 followed by a strong recovery and consistent growth in 2020 and 2021. Key observations include:

- Clear seasonality, with peak sales occurring in Q4 (September–December)
- Strong regional concentration, where West and East regions contribute the highest share of revenue
- High customer retention, with repeat customers forming the majority of the customer base
- Product performance imbalance, where a subset of products consistently generates losses despite sales volume

Using the most recent year-over-year growth trend, a forward-looking sales forecast was developed to support planning and budgeting decisions.

Final Conclusion:

The analysis demonstrates that Superstore has a scalable and growth-oriented business model, supported by strong customer retention and regional performance.

However, the presence of loss-making products highlights the need for:

- Product portfolio optimization
- Pricing and discount strategy review
- Region-specific performance management

With targeted improvements, Superstore has the potential to outperform the projected forecast and achieve sustainable long-term growth.